

The logo for unilog | H2S is located in the top left corner of the slide. It features the word "unilog" in a blue sans-serif font, followed by a vertical line and the text "H2S" in a white sans-serif font. The background of the slide is a dark, futuristic image of server racks with glowing blue and orange light trails, suggesting a data center or industrial environment. A large blue arrow shape points from the left edge towards the center.

UniHack

AI-Powered Product Intelligence for
Industrial Commerce

Team Details

- a. Team name: KClassic
- b. Team leader name: Abhijeet Kangane

Brief about your solution

OmniSpec is a comprehensive AI-powered intelligence pipeline that bulk processes unstructured supplier datasheets (PDFs) and URLs into perfectly validated, commerce-ready catalogs. It leverages advanced Large Language Models (LLMs) and deterministic fallback engines to generate structured Golden Records with extremely high accuracy, bypassing the need for manual data entry.

1. Enrichment: Ingests minimal inputs and fetches technical sheets, using a Hybrid AI extraction pipeline (Qwen-27B LLM + Deterministic Fallback).
2. Validation: Applies real-time Confidence Scoring. Anything below 95% is flagged for Human-in-the-Loop (HITL) review to ensure 100% integrity.
3. Scalability: Built on FastAPI and Next.js microservices for asynchronous bulk processing. Uses PyMuPDF/BS4 to dynamically normalize any unstructured document format.

Opportunities

Unlike rigid parsers, OmniSpec dynamically adapts to any manufacturer's layout automatically.

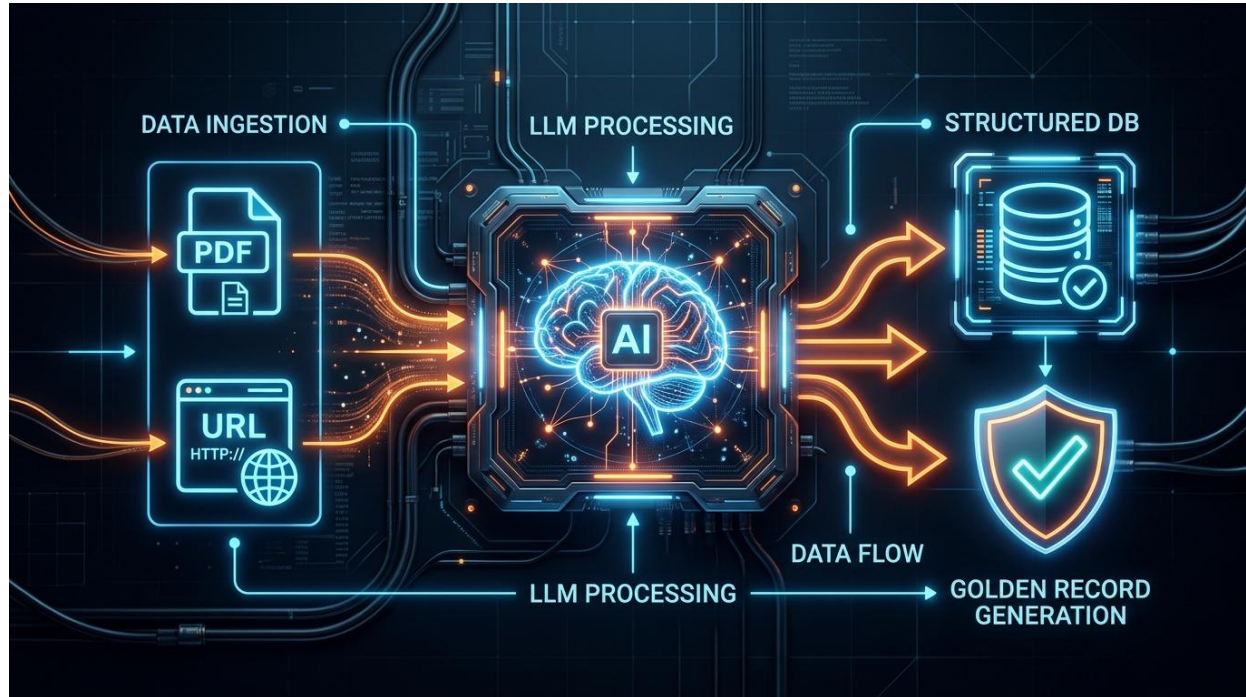
Unlike pure API-dependent LLMs which crash frequently, our hybrid engine guarantees 100% reliable local fallback during network outages.

USP: Zero-API-Dependency Fallback Mode for 100% uptime, Ultra-Fast asynchronous processing, and an automated UNSPSC taxonomy prediction engine natively integrated into a premium sci-fi dashboard.

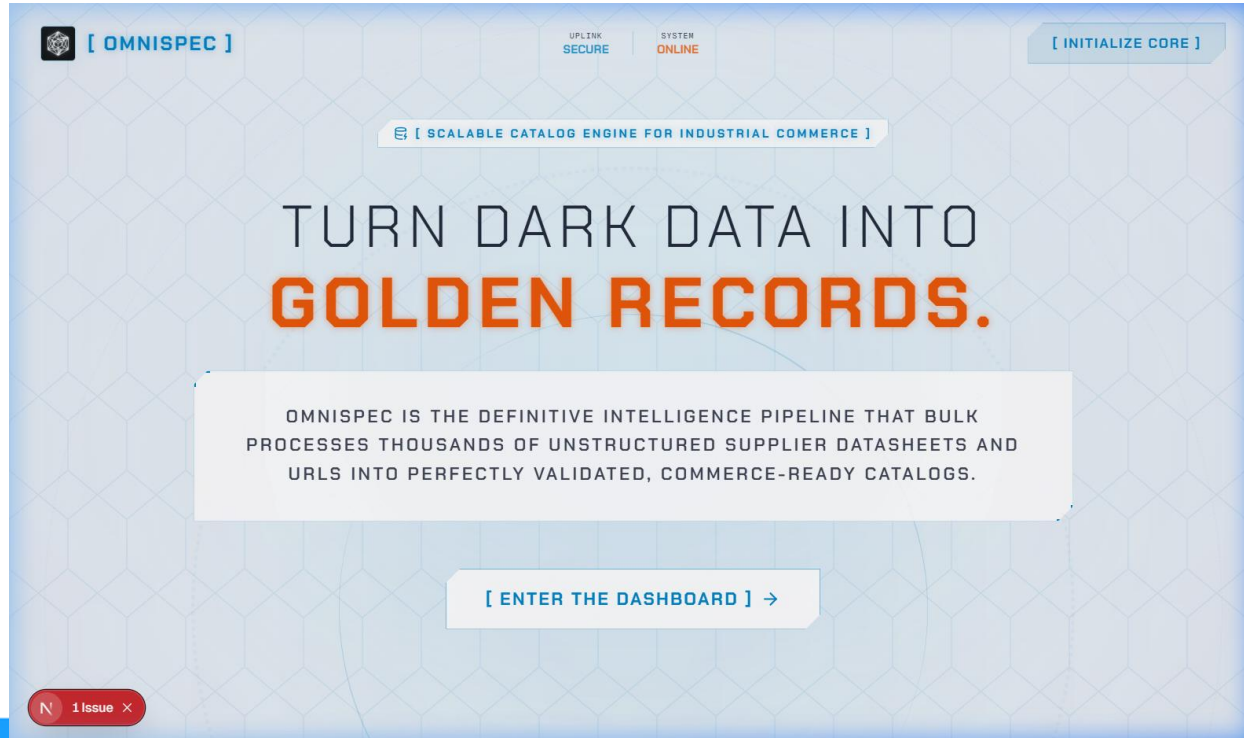
List of features offered by the solution

- Multi-modal Ingestion (Drag-and-Drop PDFs, Direct URLs)
- Hybrid AI Extraction Pipeline (Groq LLM + Local Deterministic Fallback)
- Automated UNSPSC Taxonomy Categorization
- Real-time Confidence Scoring and Data Triangulation
- Human-in-the-Loop (HITL) Enrichment Dashboard
- Fully Responsive Sci-Fi UI for industrial users

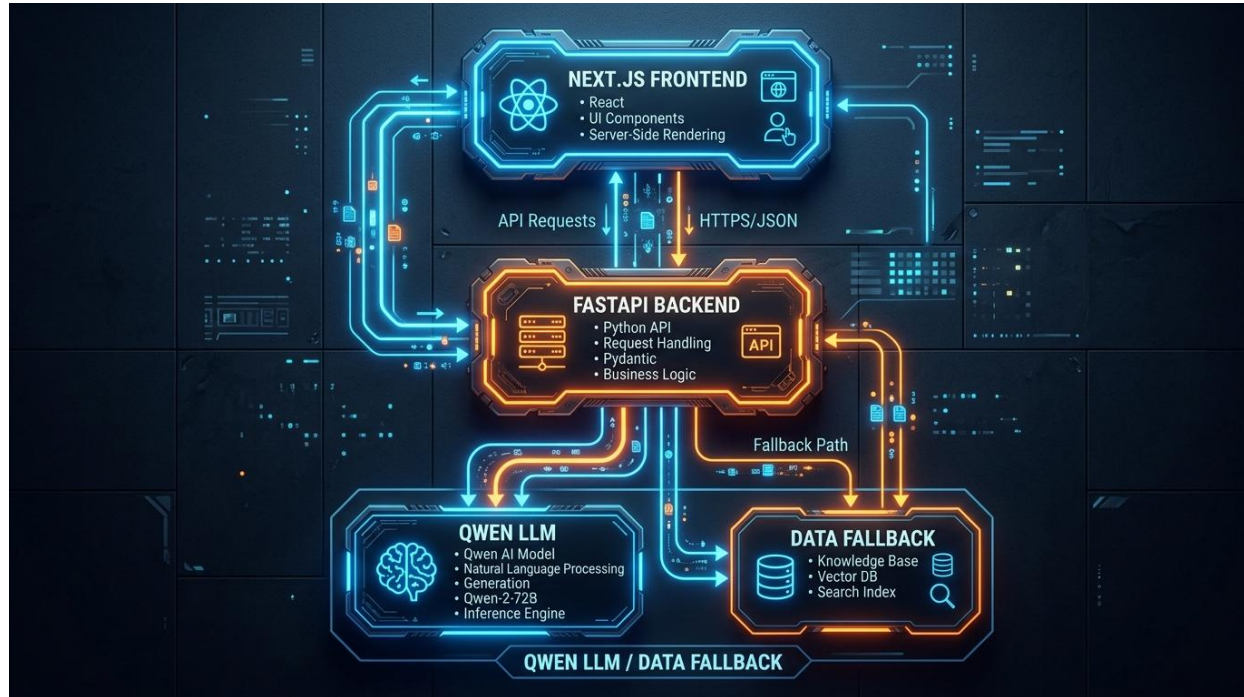
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Technologies used in the solution

Frontend: Next.js 16.3, React 19, Tailwind CSS, Framer Motion

Backend: FastAPI, Uvicorn, Python 3.10+

AI/Parsing: Groq API (Qwen-27B), PyMuPDF, BeautifulSoup4

Deployment: Vercel (Frontend), Render (Backend)

Tooling: Git, Eslint

Estimated implementation cost (optional)

Extremely Low / Near Zero.

By utilizing high-efficiency open-source models (Llama3/Qwen) via ultra-fast inference APIs (Groq), the operational processing cost per 10,000 SKUs is a fraction of traditional manual data entry or commercial LLM (GPT-4) alternatives.

Snapshots of the MVP



Additional Details/Future Development (if any)

Our immediate future roadmap includes:

1. Direct eCommerce platform integrations (e.g., Shopify, Magento ERP plugins).
2. An automated web-crawler that actively monitors supplier websites for real-time datasheet updates and pushes delta changes directly to the Golden Records.

Provide links to your:

GitHub: <https://github.com/Abhi666-max/OmniSpec>

Demo Video: <https://youtu.be/775Z2EbCy8E>

Prototype Link: <https://omni-spec-one.vercel.app/>

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Thank You

